

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ac1855df860a67f2b.jsonl`  
- SHA-256: `087923c5042294e67fe9f940b9809a506dd5bc05112ad432e9f5b2efb377c413`  
- Source modified: `2026-06-16T03:19:04+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:17:48.903Z)

Web Searcher: Single-stage TAL actionness + boundary heads, small data

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Your angle: **Single-stage TAL actionness + boundary heads, small data** ? Dimension 1 deeper/newer cut: recent single-stage TAL detectors with explicit boundary/actionness heads and whether they transfer to a 1-D trajectory + auxiliary-cue regime on tiny data.
Search query: `ActionFormer TriDet TemporalMaxer temporal action localization actionness boundary head limited training data 2023 2024 arXiv`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-16T03:17:52.610Z)

I'll search for recent single-stage TAL detectors with boundary/actionness heads and their applicability to small-data regimes.

3. user (2026-06-16T03:18:03.752Z)

Web search results for query: "ActionFormer TriDet TemporalMaxer temporal action localization actionness boundary head limited training data 2023 2024 arXiv"

Links: [{"title": "A Study in Temporal Action Localization", "url": "https://repository.tudelft.nl/file/File_7430e57c-b269-4010-a020-376bf7180e18"}, {"title": "Harnessing Temporal Causality for Advanced Temporal Action Detection", "url": "https://arxiv.org/html/2407.17792"}, {"title": "Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation/README.md at master · zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation", "url": "https://github.com/zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation/blob/master/README.md"}, {"title": "TBT-Former: Learning Temporal"}]

Boundary Distributions for Action

Localization", "url": "https://arxiv.org/pdf/2512.01298"}, {"title": "GitHub - zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation: Temporal Action Detection & Weakly Supervised Temporal Action Detection & Temporal Action Proposal Generation · GitHub", "url": "https://github.com/zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation"}, {"title": "Zero-shot Action Localization via the Confidence of Large Vision-Language Models", "url": "https://arxiv.org/pdf/2410.14340"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | Request PDF", "url": "https://www.researchgate.net/publication/373307731_TriDet_Temporal_Action_Detection_with_Relative_Boundary_Modeling"}, {"title": "Harnessing Temporal Causality for Advanced Temporal Action Detection", "url": "https://arxiv.org/pdf/2407.17792"}]

Based on the web search results, I found information about the methods you're asking about:

ActionFormer, TemporalMaxer, TriDet are recent one-stage frameworks for temporal action localization. These are all contemporary methods in the field:

- ActionFormer: Localizing Moments of Actions with Transformers was published at ECCV 2022
- TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization was published on arXiv 2023
- TriDet: Temporal action detection with relative boundary modeling was published in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition in 2023

Temporal Action Detection can be classified into one-stage, two-stage, and DETR-based methods. One-stage methods directly localize actions from multi-scale feature pyramids, and recent work has primarily focused on one-stage frameworks such as ActionFormer, TemporalMaxer, TriDet.

However, the search results don't contain specific information about actionness boundary heads or methods specifically designed for limited training data scenarios. If you need more detailed information about those specific aspects, I could perform an additional search.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

4. assistant (2026-06-16T03:18:06.779Z)

Let me refine with a couple more targeted searches to find the boundary-head specifics and the relative-boundary modeling details for TriDet, plus the TemporalMaxer simplicity angle (which matters for small data).

5. user (2026-06-16T03:18:16.602Z)

Web search results for query: "ActionFormer transformer temporal action localization single-stage regression boundary multi-scale feature pyramid ECCV 2022"

Links: [{"title": "ActionFormer: Localizing Moments of Actions with Transformers | Computer Vision ? ECCV 2022", "url": "https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29"}, {"title": "GitHub - yafeng19/actionformer: Code release for ActionFormer (ECCV 2022)", "url": "https://github.com/yafeng19/actionformer"}, {"title": "GitHub - BharathChalla/ActionFormer: ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/BharathChalla/ActionFormer"}, {"title": "GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/happyharrycn/actionformer_release"}, {"title": "GitHub - chrisindris/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/chrisindris/actionformer_release"}, {"title": "actionformer_release/README.md at main · happyharrycn/actionformer_release", "url": "https://github.com/happyharrycn/actionformer_release/blob/main/README.md"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/364819471_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/pdf/2512.01298"}]

I found comprehensive information about ActionFormer from ECCV 2022. Here are the key details:

ActionFormer: Temporal Action Localization with Transformers

ActionFormer is a model to identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows.

Technical Approach

ActionFormer combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries. More specifically, the method adapts local self-attention to model temporal context in untrimmed videos, classifies every moment in an input video, and regresses their corresponding action boundaries.

Performance Results

ActionFormer achieves 71.0% mAP at tIoU = 0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points. ActionFormer demonstrates strong results on ActivityNet 1.3 (36.6% average mAP) and EPIC-Kitchens 100 (+13.5% average mAP over prior works).

Publication Details

The work was presented at ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23-27, 2022.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. user (2026-06-16T03:18:17.112Z)

Web search results for query: "TriDet relative boundary modeling Trident-head temporal action detection CVPR 2023 boundary blur estimation"

Links: [{"title": "TriDet: Temporal Action Detection", "url": "https://www.emergentmind.com/papers/2303.07347"}, {"title": "TriDet - CVPR 2023 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2023/html/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.html"}, {"title": "CVPR Poster TriDet: Temporal Action Detection With Relative Boundary Modeling", "url": "https://cvpr2023.thecvf.com/virtual/2023/poster/21956"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/10203543/"}, {"title": "[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/abs/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2023arXiv230307347S/abstract"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/pdf/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"}, {"title": "[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"}, {"title": "GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://github.com/dingfengshi/TriDet"}]

Based on the search results, here's information about TriDet:

Overview

TriDet is a one-stage framework for temporal action detection. The paper was accepted for CVPR2023.

Key Innovation: Trident-head with Relative Boundary Modeling

Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, the paper proposes a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary.

The Trident-head consists of three components: a start head, an end head, and a center-offset head, which are designed to locate the start boundary, end boundary, and the temporal center of the action, respectively.

Feature Pyramid: Scalable-Granularity Perception (SGP)

In the feature pyramid of TriDet, a Scalable-Granularity Perception (SGP) layer is proposed to aggregate information across different temporal granularities, which is much more efficient than recent transformer-based feature pyramids.

Performance Results

TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, but with only 74.6% of its latency. TriDet achieves state-of-the-art performance on three challenging benchmarks: THUMOS14, HACS and EPIC-KITCHEN 100, with lower computational costs.

The code is available on GitHub at <https://github.com/dingfengshi/TriDet>.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-06-16T03:18:20.360Z)

Let me do two more targeted searches: one on TemporalMaxer (max-pooling simplicity, relevant for small data), and one on action-boundary regression / over-segmentation for action segmentation (ASRF/MS-TCN++) since that's the most directly transferable to the over-merge problem.

8. user (2026-06-16T03:18:31.221Z)

Web search results for query: "ASRF action boundary regression smoothing over-segmentation MS-TCN++ temporal action segmentation segmental F1 edit score"

Links: [{"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "Exploring Ordinal Bias in Action Recognition for Instructional Videos", "url": "https://arxiv.org/html/2504.06580"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "(PDF) MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://www.researchgate.net/publication/344127554_MS-TCN_Multi-Stage_Temporal_Convolutional_Network_for_Action_Segmentation"}, {"title": "Exploring Ordinal Bias in Action Recognition for Instructional Videos", "url": "https://arxiv.org/pdf/2504.06580"}, {"title": "(PDF) Boundary-Centric Active Learning for Temporal Action Segmentation", "url": "https://www.researchgate.net/publication/403906082_Boundary-Centric_Active_Learning_for_Temporal_Action_Segmentation"}, {"title": "Boundary-Centric Active Learning for Temporal Action Segmentation", "url": "https://arxiv.org/html/2604.15173v1"}, {"title": "Combining Boundary Supervision and Segment-Level Regularization for Fine-Grained Action Segmentation", "url": "https://arxiv.org/html/2604.01859"}]

Based on the search results, here's what I found about these temporal action segmentation approaches:

ASRF (Action Segmentation and Regression Framework)

ASRF proposes a framework for temporal action segmentation that divides the task into frame-wise action classification and action boundary regression. ASRF improves segmentation quality by adding an auxiliary branch that explicitly regresses action boundaries to mitigate over-segmentation errors.

MS-TCN and MS-TCN++

MS-TCN employs a multi-stage architecture with dilated temporal convolutions and a smoothing loss to iteratively refine frame-level predictions. MS-TCN++ extends this approach by integrating dual dilated layers that capture both local and global contexts while decoupling prediction generation from refinement.

Addressing Over-Segmentation

Smoothing losses (truncated MSE on log probabilities) enforce temporal coherence, addressing the over-segmentation problem. Over-segmentation manifests as boundary jitter and fragmentation into short sub-segments and disproportionately degrades segment-level metrics.

Evaluation Metrics

While the search results reference segmental F1 and edit scores as standard evaluation metrics for temporal action segmentation that explicitly penalize over-segmentation and boundary errors, the results don't provide detailed information about how these specific metrics are calculated.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-06-16T03:18:32.310Z)

Web search results for query: "TemporalMaxer maximize temporal context max pooling temporal action localization 2023 arXiv parameter-free"

Links: [{"title": "GitHub - TuanTNG/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization", "url": "https://github.com/TuanTNG/TemporalMaxer"}, {"title": "GitHub - f1brahim-tmu/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization · GitHub", "url": "https://github.com/f1brahim-tmu/TemporalMaxer"}, {"title": "A Study in Temporal Action Localization", "url": "https://repository.tudelft.nl/file/File_7430e57c-b269-4010-a020-376bf7180e18"}, {"title": "TemporalMaxer/README.md at master · TuanTNG/TemporalMaxer", "url": "https://github.com/TuanTNG/TemporalMaxer/blob/master/README.md"}, {"title": "Temporal Max Pooling for Sequence Modeling", "url": "https://www.emergentmind.com/topics/temporal-max-pooling"}, {"title": "Temporal Action Localization with Cross Layer Task Decoupling and Refinement", "url": "https://arxiv.org/html/2412.09202v1"}, {"title": "[2303.09055] TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization", "url": "https://arxiv.org/abs/2303.09055"}, {"title": "TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization | Papers With Code", "url": "https://paperswithcode.com/paper/temporalmaxer-maximize-temporal-context-with"}, {"title": "TemporalMaxer: Maximize Temporal Context with only Max Pooling", "url": "https://arxiv.org/pdf/2303.09055"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/pdf/2512.01298"}]

I found information about TemporalMaxer! Here are the key results:

Overview

TemporalMaxer minimizes long-term temporal context modeling while maximizing information from extracted video clip features with a basic, parameter-free, and local region operating max-pooling block.

Publication Details

The paper was authored by Tang, Tuan N, Kim, Kwonyoung, and Sohn, Kwanghoon, published on arXiv as arXiv:2303.09055 in 2023.

Task and Motivation

Temporal Action Localization (TAL) is a challenging task in video understanding that aims to identify and localize actions within a video sequence. Recent studies have emphasized the importance of applying long-term temporal context modeling (TCM) blocks to extracted video clip features such as employing complex self-attention mechanisms.

Key Approach and Results

The paper presents the simplest method to address this task, arguing that extracted video clip features are already informative to achieve outstanding performance without sophisticated architectures. The method picks out only the most critical information for adjacent and local clip embeddings, resulting in a more efficient TAL model.

Empirical results show TemporalMaxer surpasses transformer-based TCM blocks by 0.9?1.3 mAP across THUMOS14, EPIC-Kitchens100, MultiTHUMOS, and MUSES.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-06-16T03:18:35.734Z)

I have strong coverage of the single-stage TAL detectors with explicit boundary/actionness heads (ActionFormer, TriDet, TemporalMaxer) plus the directly-transferable boundary-regression-for-over-segmentation method (ASRF) and the segmentation backbone (MS-TCN++). Let me return the ranked results.

11. user (2026-06-16T03:19:02.279Z)

Structured output provided successfully

12. assistant (2026-06-16T03:19:04.218Z)

I've completed the search and returned the ranked results via StructuredOutput.